Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$\frac{-1}{6x}+c$
			Total 1 mark

Q2.

Marking Instructions	AO	Marks	Typical Solution
Expands the bracket and obtains any correct form (PI)	AO1.1b	B1	$f'(x)=4x^2-12+\frac{9}{x^2}$
Integrates at least one of 'their' terms correctly	AO1.1a	M1	$f(x)=\int 4x^2-12+\frac{9}{x^2}\,dx$ $=\frac{4x^3}{3}-12x-\frac{9}{x}+c$
Substitutes x = 3 into 'their' integrated expression and equates to 2	AO1.1a	M1	f(3) = 36 − 36 − 3 + c = 2 c = 5
Obtains completely correct expression, must be explicitly stated CAO (ACF)	AO1.1b	A1	$f(x)=\frac{4x^3}{3}-12x-\frac{9}{x}+5$
			Total 4 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Writes in a form to which the binomial expansion can be applied Accept $A\left(1+\frac{x}{4}\right)^{-\frac{1}{2}}$	AO3.1a	M1	$\frac{1}{\sqrt{4+x}}=\frac{1}{2}\left(1+\frac{x}{4}\right)^{-\frac{1}{2}}$ $\approx\frac{1}{2}\left[1+\left(-\frac{1}{2}\right)\frac{x}{4}+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(\frac{x}{4}\right)^2}{2!}\right]$
	Uses binomial expansion for their $(1+kx)^{\pm\frac{1}{2}}$ with at least two terms correct (can be unsimplified)	AO1.1a	M1	$\approx\frac{1}{2}\left[1-\frac{x}{8}+\frac{3x^2}{128}\right]$ $\approx\frac{1}{2}-\frac{1}{16}x+\frac{3}{256}x^2$

	Marking Instructions	AO	Marks	Typical Solution
	Obtains correct simplified answer No need to expand brackets CAO	AO1.1b	A1	
(b)	Substitutes $-x^3$ in their three term expansion from part (a)	AO1.1a	M1	$\frac{1}{\sqrt{4-x^3}} \approx \frac{1}{2} - \frac{1}{16}(-x^3) + \frac{3}{256}(-x^3)^2$
	Obtains correct expansion. FT their (a)	AO1.1b	A1F	$\approx \frac{1}{2} + \frac{x^3}{16} + \frac{3x^6}{256}$
(c)	Uses their three term expansion as the integrand ignore limits PI by next mark	AO1.1a	M1	$\int_0^1 \frac{1}{\sqrt{4-x^3}} \mathrm{d}x \approx \int_0^1 \frac{1}{2} + \frac{x^3}{16} + \frac{3x^6}{256} \mathrm{d}x$
	Integrates (at least two terms correct)	AO1.1a	M1	$\approx \left[\frac{x}{2} + \frac{x^4}{64} + \frac{3x^7}{1792}\right]_0^1$
	Obtains correct value CAO	AO1.1b	A1	$\approx \frac{1}{2} + \frac{1}{64} + \frac{3}{1792}$ ≈ 0.5172991
(d)(i)	Explains that each term in the expansion is positive	AO2.4	E1	Each term in the expansion is positive.
	Deduces that increasing the number of terms will increase the estimated value and that the value must be an underestimate. (Condone inference if evidence given ie value calculated numerically and compared)	AO2.2a	R1	So increasing the terms will increase the estimated value hence the value must be an underestimate.
(ii)	States the validity of their binomial expansion for part (b) Provided their $k \neq \pm 1$	AO3.1a	B1F	The binomial expansion is valid for $\|x\| < \sqrt[3]{4}$
	Compares integral lower limit with validity of correct expansion CAO	AO2.3	E1	$2 > \sqrt[3]{4}$
				Total 12 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates, with at least one term correct	AO1.1a	M1	$\frac{\mathrm{d}v}{\mathrm{d}t} = 12t - 36^2$

	Selects and applies $F = ma$ to 'their' derivative Condone use of 400 for mass	AO1.1a	M1	$F = ma = 0.4\ (12t - 36t^2)$
	Obtains correct expression for force FT from 'their' $F = ma$ equation, provided the first M1 has been awarded (may be in factorised form)	AO1.1b	A1F	$= 4.8t - 14.4t^2$
(b)	Integrates v to find r, with at least one term correct	AO3.1b	M1	$r = \int (6t^2 - 12t^3)\,dt$
	Obtains correct integral (condone absence of c)	AO1.1b	A1	$r = 2t^3 - 3t^4 + c$
	Deduces the value of c using initial conditions FT use of 'their' integral provided M1 awarded	AO2.2a	A1F	When $t = 0$, $r = 0$ so $0 = 2 \times 0^2 - 3 \times 0^4 + c$ so $c = 0$
	Forms and solves an equation for t (condone numerical slip)	AO1.1a	M1	$0 = 2t^3 - 3t^4$ $= t^3(2 - 3t)$ $t = 0$ or $t = \frac{2}{3}$
	Interprets solution, realising that the non-zero time is required (Must include units) FT use of 'their' equation for t provided both M1 marks have been awarded	AO3.2a	A1F	Next at O at $\frac{2}{3}$ seconds
				Total 8 marks

Q5.

Marking Instructions	AO	Marks	Typical Solution
States a correct integral expression (ignore limits at this stage)	AO1.1a	M1	$\text{Area} = \int_a^{2a} \left(6x^2 + \frac{8}{x^2}\right) dx$
Integrates at least one term correctly	AO1.1b	A1	$= \left[2x^3 - \frac{8}{x}\right]_a^{2a}$

Substitutes $2a$ and a into 'their' integrated expression	AO1.1a	M1	$= (16a^3 - \frac{4}{a}) - (2a^3 - \frac{8}{a})$
States correct final answer with terms collected FT correct substitution into incorrect integral provided both M1 marks awarded	AO1.1b	A1F	$= 14a^3 + \frac{4}{a}$
			Total 4 marks

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Separates variables, at least one side correct.	AO3.1a	M1	$3\sqrt{x}\frac{dx}{dt} = 8\sin 2t$
	Obtains correct separation PI	AO1.1b	A1	$\int 3\sqrt{x}\,dx = \int 8\sin 2t\,dt$
	integrates 'their' expressions, at least one of 'their' sides correct	AO1.1a	M1	$\int 3x^{\frac{1}{2}}dx = \int 8\sin 2t\,dt$ $2x^{\frac{3}{2}} = -4\cos 2t(+c)$
	Obtains correct integral (condone missing + c) CAO	AO1.1b	A1	
	Substitutes initial conditions, to find + c.	AO3.1b	M1	$2 \times (0)^{\frac{3}{2}} = -4\cos(2 \times 0) + c$ $c = 4$
	Obtains a correct solution ACF	AO1.1b	A1	$x^{\frac{3}{2}} = 2 - 2\cos 2t$
	Obtains correct solution of the form $x = f(t)$	AO2.5	A1	$x = (2 - 2\cos 2t)^{\frac{2}{3}}$
(b)	Obtains correct max height, in cm Award FT from correct substitution into incorrect equation $x = f(t)$ but only if all three M1 marks have been awarded, must have correct units.	AO3.4	A1F	max height $= 4^{\frac{2}{3}} = 252$ cm
				Total 8 marks

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Integrates both components with at least one correct	AO1.1a	M1	$\mathbf{r} = \int 40e^{-0.2t}\,dt\,\mathbf{i} + \int 50(e^{-0.2t} - 1)\,dt\,\mathbf{j}$ $= (-200e^{-0.2t} + c)\mathbf{i} + (-250e^{-0.2t} - 50t + d)\mathbf{j}$
	Obtains correct terms. (condone missing constants)	AO1.1b	A1	$t = 0$, $\mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow c = 200$, $d = 250$ **OR**
	Evaluates both constants (or uses definite integration) using 'their' expression for **r**	AO3.4	M1	$\mathbf{r} = \int_0^t 40e^{-0.2t}\,dt\,\mathbf{i} + \int_0^t 50(e^{-0.2t} - 1)\,dt\,\mathbf{j}$ $= \left[-200e^{-0.2t}\mathbf{i} + (-250e^{-0.2t} - 50t)\mathbf{j}\right]_0^t$
	Obtains correct expression	AO1.1b	A1	$\mathbf{r} = 200(1 - e^{-0.2t})\mathbf{i}$ $+ (250 - 250e^{-0.2t} - 50t)\mathbf{j}$
(b)	Forms equation to find t based on horizontal component	AO3.4	M1	$200(1 - e^{-0.2t}) = 100$ $t = 5\ln 2 = 3.4657$ $y = 250 - 250e^{-0.2 \times 5\ln 2} - 50 \times 5\ln 2$ $= -48.3$
	Obtains correct time	AO1.1b	A1	
	Substitutes 'their' time into vertical component	AO1.1a	M1	
	Obtains correct displacement for 'their' time (to 1 sf, must have metres) FT only if both M1 marks have been awarded	AO3.2a	A1F	The parachutist has a vertical displacement of 50 m below the origin
(c)	Identifies vertical component of the velocity as first step	AO2.4	M1	$\frac{d}{dt}(50(e^{-0.2t} - 1)) = -10e^{-0.2t}$ As there is no initial vertical component of velocity (and hence no air resistance) the initial acceleration is only due to gravity. Hence g is taken as 10 m s^{-2}
	Differentiates vertical component of velocity correctly	AO1.1b	A1	
	Considers implication of initial motion and reaches correct conclusion	AO2.2a	R1	
				Total 11 marks

Q8.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	18
			Total 1 mark

Q9.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving $\frac{dA}{dt}$, A and a constant of proportionality.	AO3.3	M1	$\frac{dA}{dt} \propto A$ $\Rightarrow \frac{dA}{dt} = kA$ $\Rightarrow \int \frac{1}{A} dA = \int k\,dt$ $\Rightarrow \ln A = kt + c$ $\Rightarrow A = e^{kt+c}$ $\Rightarrow A = Be^{kt}$ **AG**
	Separates variables	AO1.1a	M1	
	Integrates both of 'their' sides correctly	AO1.1b	A1F	
	Constructs a rigorous mathematical argument that supports use of the given model. **AG** Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
(b)(i)	States correct value of B	AO1.1b	B1	$B = 0.25$ or $B = \frac{1}{4}$
(ii)	Uses $t = 20$ and $A = 0.5$ to find k	AO3.1b	M1	When $t = 20$, $A = 0.5$ $\Rightarrow 0.5 = 0.25e^{20k}$ $\Rightarrow 20k = \ln 2$ $\Rightarrow k = \frac{1}{20}\ln 2$ $\Rightarrow A = \frac{1}{4}(e^{\ln 2})^{\frac{t}{20}}$ $\Rightarrow A = 2^{-2} \times 2^{\frac{t}{20}}$ $A = 2^{\frac{t}{20}-2}$ **AG**
	Finds correct value of k	AO1.1b	A1	
	Substitutes 'their' k to get A in terms of t	AO1.1a	M1	
	Constructs rigorous and convincing argument to show $A = 2^{\frac{t}{20}-2}$ Using correct notation	AO2.1	R1	

	throughout. **AG**			
(iii)	Uses the model to set up correct equation and attempt to find t	AO3.4	M1	$2\pi = 2^{\frac{t}{20}-2}$ t = 93.03 days
	Finds correct value of t	AO1.1b	A1	

(c)	States any sensible and relevant limitation of the model that is specified in terms of the pond, area, weed, rate of change or time.	AO3.5b	E1	Model predicts that the area of weed will increase without limit and this is not possible since the area of the pond is 4π
(d)	Any sensible and relevant refinement to the model that is specified in terms of the pond, area, weed, rate of change or time	AO3.5c	E1	Introduce a limiting factor such as fish eating weed or rate of growth decreases as surface area covered
				Total 13 marks

Q10.

Marking Instructions	AO	Marks	Typical Solution
Selects a method of integration, which could lead to a correct solution. Evidence of integration by parts OR an attempt at integration by inspection.	AO3.1a	M1	$u = \ln 2x,\quad \frac{dv}{dx} = x^3$ $\frac{du}{dx} = \frac{1}{x};\quad v = \frac{x^4}{4}$ $\left[\frac{x^4}{4}\ln(2x)\right]_1^2 - \int_1^2 \frac{x^3}{4}dx$
Applies integration by parts formula correctly OR correctly differentiates an expression of the form $Ax^4\ln 2x$	AO1.1b	A1	$\left[\frac{x^4}{4}\ln(2x) - \frac{x^4}{16}\right]_1^2$ $= \left(\frac{2^4}{4}\ln(4) - \frac{2^4}{16}\right) - \left(\frac{1}{4}\ln(2) - \frac{1}{16}\right)$
Obtains correct integral, condone missing limits.	AO1.1b	A1	
Substitutes correct limits into 'their' integral	AO1.1a	M1	$\frac{31}{4}\ln 2 - \frac{15}{16}$
Obtains correct p and q FT use of incorrect integral provided both M1 marks	AO1.1b	A1F	so $p = \frac{31}{4}\quad q = -\frac{15}{16}$ **ALT**

have been awarded			$\frac{d}{dx}(x^4\ln 2x) = 4x^3\ln 2x + x^4 \cdot \frac{1}{x}$ $\therefore \int_1^2 x^3\ln 2x\,dx = \left[\frac{1}{4}(x^4\ln 2x - \frac{x^4}{4})\right]_1^2$ $= \left(\frac{2^4}{4}\ln(4) - \frac{2^4}{16}\right) - \left(\frac{1}{4}\ln(2) - \frac{1}{16}\right)$ $\frac{31}{4}\ln 2 - \frac{15}{16}$ $p = \frac{31}{4} \quad q = -\frac{15}{16}$
			Total 5 marks

Q11.

(a) $h = 1$

PI

B1

$f(x) = \frac{1}{x^2+1}$

$I \approx \frac{h}{2}$ {f(1) + f(5) + 2[f(2) + f(3) + f(4)]}

$\frac{h}{2}$ {f(1) + f(5) + 2[f(2) + f(3) + f(4)]}

OE summing of areas of the four ‘trapezia’…

M1

$\frac{h}{2}$ with $\{\ldots\} = \frac{1}{2} + \frac{1}{26} + 2\left(\frac{1}{5} + \frac{1}{10} + \frac{1}{17}\right)$

= 0.5 + 0.03(84...) + 2[0.2+0.1+0.05(88...)]
= 0.538(46…) + 2[0.358(82...)] = 1.256(108…)

OE Accept 2dp (rounded or truncated) for non-terminating decs. equiv.

A1

($I \approx$) 0.628054… = $\frac{694}{1105}$ = 0.628 (to 3sf)

CAO Must be 0.628

***SC** for those who use 5 strips, max possible is B0M1A1A0*

A1

4

(b) (i) $\int\left(x^{-\frac{3}{2}}+6x^{\frac{1}{2}}\right)dx=\frac{x^{-\frac{1}{2}}}{-1/2}+\frac{6x^{\frac{3}{2}}}{3/2}$ (+ c)

One term correct (even unsimplified)

M1

Both terms correct (even unsimplified)

A1

$= -2x^{-0.5} + 4x^{1.5}$ (+ c)

Must be simplified

A1

3

(ii) $\int_1^4\left(x^{-\frac{3}{2}}+6x^{\frac{1}{2}}\right)$ $dx = [-2(4^{-0.5}) + 4(4^{1.5})] - [-2(1^{-0.5}) + 4(1^{1.5})]$

Attempt to calculate F(4) − F(1) where F(x) follows integration and is not just the integrand

M1

$= (-1 + 32) - (-2 + 4) = 29$

Since 'Hence' NMS scores 0 / 2

A1

2

[9]

Q12.

(a) $\int e^{1-2x}\,dx = ke^{1-2x}$ or $e(ke^{-2x})$

where k is a rational number

M1

$\int_0^{\ln 2} e^{1-2x}dx = -\frac{1}{2}e^{1-2x}\Big|_0^{\ln 2}$ or $e\left[-\frac{1}{2}e^{-2x}\right]_0^{\ln 2}$

correct integration

condone missing limits

A1

$= -\frac{1}{2}e^{1-2\ln 2} - -\frac{1}{2}e^{1-2(0)}$

correct (no decimals)

A1

$= -\frac{1}{2}\left(\frac{1}{4}e\right) + \frac{1}{2}e$

eliminating ln

$= \frac{3}{8}e$

AG, be convinced

A1

4

(b) $u = \tan x$

$\frac{du}{dx} = \sec^2 x$

PI below, condone $du = \sec^2 x dx$

M1

Replacing dx by $\frac{1}{\sec^2 x}(du)$ in integral

or $\frac{1}{1+u^2}(du)$

A1

$\sec^2 x = 1 + u^2$

PI below

B1

$$\left.\begin{array}{l} x = 0 \Rightarrow u = 0 \\ x = \frac{\pi}{4} \Rightarrow u = 1 \end{array}\right\}$$

this could be gained by changing u to $\tan x$ after the integration and using $x = 0$ and $x = \frac{\pi}{4}$

B1

$$\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt{\tan x}\, dx$$

$= \int (1+u^2)\sqrt{u}\,(du)$ or $\int (1+u^2)^2 \sqrt{u} \frac{(du)}{1+u^2}$

all in terms of u including replacing dx

all correct, condone omission of du

M1

$= \int\left(u^{\frac{5}{2}} + u^{\frac{1}{2}}\right) \ (\mathrm{d}u)$

must be in this form

A1

$= \frac{2}{7}u^{\frac{7}{2}} + \frac{2}{3}u^{\frac{3}{2}}$

accept correct unsimplified form

A1

$= \frac{20}{21}$

CAO

A1 **8**

[12]

Q13.

(a) $\int x(x^2+3)^{\frac{1}{2}} \, \mathrm{d}x = p(x^2+3)^{\frac{3}{2}}$

By inspection or substitution

M1

$= \frac{1}{3}(x^2+3)^{\frac{3}{2}} \ (+C)$

A1 **2**

(b) $\int \mathrm{e}^{2y} \, \mathrm{d}y = \int x\sqrt{x^2+3} \, \mathrm{d}x$

Correct separation and notation
Condone missing integral signs

B1

$= \frac{1}{2}\mathrm{e}^{2y}$

B1

$= \frac{1}{3}(x^2+3)^{\frac{3}{2}} + C$

Equate to result from (a) with constant.

M1

$\frac{1}{2} = \frac{1}{3} \times 4^{\frac{3}{2}} + C$

Use (1,0) to find constant.

m1

$C = -\frac{13}{6}$

CAO

A1

$2y = \ln\left(\frac{2}{3}(x^2+3)^{\frac{3}{2}} - \frac{13}{3}\right)$

Solve for y, taking logs correctly.

m1

$y = \frac{1}{2}\ln\left(\frac{2}{3}(x^2+3)^{\frac{3}{2}} - \frac{13}{3}\right)$

CSO

A1 7

[9]

Q14.

(a) $\frac{dy}{dx} = 5x^4 - 4x$

one of these terms correct

M1

all correct (no + c etc)

A1

$(= 5(-1)^4 - 4(-1))$ $= 9$

A1

Tangent has equation y = *'their 9'* $x + c$

and 6 = *'their 9'* $(-1) + c \Rightarrow = ...$

tangent using 'their□ gradient, and attempt to find c using $x = -1$ and $y = 6$

m1

$\Rightarrow y = 9x + 15$

equation must be seen in this form

A1

5

(b) (i) When $x = 2$, $y = 2^5 - 2 \times 2^2 + 9 = 32 - 8 + 9 = 33$

be convinced that they are using ***curve*** *equation*

$(k =)$ 33

NMS *k = 33 scores B0*

B1

1

(ii) When $x = 2$, $y = 9 \times 2 + 15 = 33$ so lies on tangent

be convinced that they are using ***tangent*** *equation* ***and*** *have statement*

B1

1

(c) (i) $\frac{x^6}{6} - \frac{2x^3}{3} + 9x$

one of these terms correct

M1

another term correct

A1

all correct (may have +c)

A1

$$\left[\frac{2^6}{6} - \frac{2 \times 2^3}{3} + 9 \times 2\right] - \left[\frac{(-1)^6}{6} - \frac{2 \times (-1)^3}{3} + 9 \times (-1)\right]$$

$$\left[\frac{64}{6} - \frac{16}{3} + 18\right] - \left[\frac{1}{6} + \frac{2}{3} - 9\right]$$

= 31.5

F(2) − F(−1) unsimplified FT "their terms" from integration = $\frac{70}{3} - \left(-\frac{49}{6}\right)$

m1

$\left(or\ \frac{189}{6}\ etc\right)$

condone single fraction

A1

5

(ii) Area of trapezium = $\frac{1}{2}$ × 3 × (6 + 'their' k)

= 58.5 when k = 33

B1√

Shaded area = **Trapezium** − 'their' (c)(i) value

M1

= 27

OE eg $\frac{162}{6}$

A1

3

[15]

Q15.

(a) (i) $\{(2 + y)^3 =\}\ 8 + 12y + 6y^2 + y^3$

At least 3 terms simplified and correct

M1

All correct

A1

2

(ii) $(2 + x^{-2})^3 = 8 + 12x^{-2} + 6(x^{-2})^2 + (x^{-2})^3$

A replacement of y by x^{-2} in c's (a)(i) working. PI

M1

$(2 - x^{-2})^3 = 8 - 12x^{-2} + 6(x^{-2})^2 - (x^{-2})^3$

Ft one incorrect coefficient in (a)(i) expansion.

A1F

$(2 + x^{-2})^3 + (2 - x^{-2})^3 = 16 + 12x^{-4}$

CSO Be convinced.

***SC2** for a fully correct solution, not using 'Hence'*

A1

3

(b) (i) $\int \left[(2+x^{-2})^3 + (2-x^{-2})^3\right] \mathrm{d}x = 16x - 4x^{-3}(+c)$

Valid method to obtain the correct power of x after *integrating* qx^{-4}.

M1

$16x - 4x^{-3}$ *or* $16x - 4/x^3$ *condone missing '+c'.*

Ft on c*'s* p *and* q *values. Coefficients and signs must be simplified*

A1F

2

(ii) $\int_1^2$ .... $\mathrm{d}x = [16(2) - 4(2^{-3})] - [16 - 4]$

F(2) − F(1) following integration (b)(i)

M1

$= 31.5 - 12 = 19.5$

OE Ft on c's ***positive integer*** *values of* p *and* q.

Since 'Hence' NMS scores 0/2

A1F

2

[9]

Q16.

(a) (i) $5 - 8x = A(1 - 3x) + B(2 + x)$

M1

$x = -2 \quad x = \frac{1}{3}$

Two values of x *used to find values for* A *and* B

m1

$A = 3 \quad B = 1$

A1

3

Alternative:

$5 - 8x = A(1 - 3x) + B(2 + x)$

(M1)

$5 = A + 2B$

$-8 = -3A + B$

Set up simultaneous equations and solve.

(m1)

$A = 3 \quad B = 1$

(A1) **(3)**

(ii) $\int_{-1}^{0} \frac{3}{2+x} + \frac{1}{1-3x}\, dx = 3 \ln (2 + x) - \frac{1}{3} \ln (1 - 3x)$

a ln(2 + x) + b ln(1 − 3x) where a and b are constants

M1

$= (3 \ln 2 - \frac{1}{3}\ln 1) - (3 \ln 1 - \frac{1}{3} \ln 4)$

f(0) − f(−1) used

m1

$= 3 \ln 2 + \frac{1}{3} \ln 4$

ft A and B

A1ft

$= \frac{11}{3} \ln 2$

ft $\left(A + \frac{2}{3}B\right)$ ln 2

A1ft **4**

(b) (i) $(C =)2$

B1 **1**

(ii) $\int \frac{9-18x-6x^2}{2-5x-3x^2}\, dx = \int C dx + \int \frac{5-8x}{2-5x-3x^2}\, dx$

Seen or implied.

Allow $\pm C + \int \frac{5-8x}{2-5x-3x^2} dx$

M1

$\int_{-1}^{0} \frac{9-18x-6x^2}{2-5x-3x^2}\, dx = 2 + \frac{11}{3} \ln 2$

Accept 2 + 3 ln 2 + $\frac{1}{3}$ ln 4

ft 2 + candidate's answer to part (a)(ii) if exact.

A1ft **2**

[10]

Q17.

(a) $\int t\cos\left(\frac{\pi}{4}t\right)\,dt$

Clear attempt to use parts

$u = t$ $\quad \frac{dv}{dt} = \cos\left(\frac{\pi}{4}t\right)$

$\frac{du}{dt} = 1$ $\quad v = k\sin\left(\frac{\pi}{4}t\right)$

M1

$= t \times \frac{4}{\pi}\sin\left(\frac{\pi}{4}t\right) - \frac{4}{\pi}\int \sin\left(\frac{\pi}{4}t\right)\,(dt)$

Must be in terms of π

A1

$= pt\sin\left(\frac{\pi}{4}t\right) + q\cos\left(\frac{\pi}{4}t\right)$

Correct form, any non-zero values for p, q

m1

$= t \times \frac{4}{\pi}\sin\left(\frac{\pi}{4}t\right) + \frac{4}{\pi} \times \frac{4}{\pi}\cos\left(\frac{\pi}{4}t\right)$

Any correct unsimplified form.
Constant not required

A1

4

(b) $\int 32x\,dx = \int t\cos\left(\frac{\pi}{4}t\right)\,dt$

Correct separation and notation.

B1

$16x^2 =$

$\frac{x^2}{2}$ *if 32 not brought over; allow* $32 \times \frac{x^2}{2}$

B1

$t \times \frac{4}{\pi}\sin\left(\frac{\pi}{4}t\right) + \frac{16}{\pi^2}\cos\left(\frac{\pi}{4}t\right) + C$

Equate to result from part (a) with constant and use (0,4) to find a value for the constant

M1

$C = 256 - \frac{16}{\pi^2}$

Accept C = 254 or better(254.37886...)

A1

$t = 45$

$16x^2 = -40.514... - 1.146... + 254.378... = 212.718...$
$x^2 = 13.294...$
$x = 3.646... = 3.65$ m or (height =) 365 cm

Substitute t = 45 into $kx^2 = pt\sin\left(\frac{\pi}{4}t\right) + q\cos\left(\frac{\pi}{4}t\right) + C$
$p \neq 0$, $q \neq 0$ and calculate x.
CSO

m1A1

6

[10]

Q18.

(a) $\frac{dy}{dx} = 12 - 5x^{\frac{2}{3}}$

$kx^{\frac{2}{3}}$ term.

M1

ACF

A1

2

(b) (i) When $x = 0$, $\frac{dy}{dx} = 12$

Ft on c's y' evaluated correctly at $x = 0$

B1F

Eqn of tangent at O is $y = 12x$

OE Ft on c's value for $y'(0)$ provided $y'(0) > 0$.

B1F

2

(ii) When $x = 8$, $\frac{dy}{dx} = 12 - 5 \times (8)^{\frac{2}{3}}$

Attempt to find $\frac{dy}{dx}$ when $x = 8$

M1

Equation of tangent at (8, 0) is $y - 0 = y'(8)[x - 8]$

$y = y'(8)[x - 8]$ OE

M1

$y = -8(x - 8) \Rightarrow y + 8x = 64$

CSO AG

A1

3

(c) $\int \left(12x - 3x^{\frac{5}{3}}\right) dx = \frac{12x^2}{2} - \frac{3x^{\frac{8}{3}}}{\frac{8}{3}} \ (+c)$

$kx^{\frac{5}{3}+1}$ term after integrating, condone k left unsimplified for this M mark.

$= 6x^2 - \frac{9}{8}x^{\frac{8}{3}} \ (+c)$

For $6x^2$ OE eg $(12x^2 / 2)$

B1

For $-\frac{9}{8}x^{\frac{8}{3}}$ OE

A1

3

(d) Area bounded by curve and x-axis

$= \int_0^8 \left(12x - 3x^{\frac{5}{3}}\right) dx = 6 \times 8^2 - \frac{9}{8} \times (8)^{\frac{8}{3}}$

$\pm$ F(8) {- F(0)} PI following integration

M1

= 384 − 288 = 96

PI by correct final answer if evaluation not seen here

A1

At P, $12x + 8x = 64$

Solving $y + 8x = 64$ and c's $y = kx$, $k > 0$, down to an eqn in one variable…

$[y + 2y / 3 = 64]$

M1

(x_p = 3.2) y_p = 38.4

For y_p38.4 OE [If using integration to find area of triangle, award A1 if both 'x_p = 3.2' and correct integration of correct eqns of the 2 lines]

A1

Area of triangle OPA = $\frac{1}{2}$ × 8 × y_p

OE Need perpendicular ht to be linked to y_p > 0.

M1

Area of shaded region = Area Δ OPA − $\int_0^8 \left(12x - 3x^{\frac{5}{3}}\right)dx$

M0 if evaluated to a value < 0

M1

= 153.6 − 96 = 57.6

OE eg 288 / 5

A1 7

[17]

Q19.

(a) $2x + 3 = A(2x + 1) + B(2x - 1)$

M1

$x = \frac{1}{2}$ $x = -\frac{1}{2}$

Use two values of x to find A and B

m1

$A = 2$ $B = -1$

Both

A1 3

Alternative; equating coefficients

$2x + 3 = A(2x + 1) + B(2x - 1)$

(M1)

x term $2 = 2A + 2B$
constant $3 = A - B$
Set up simultaneous equations and solve.

(m1)

$A = 2 \quad B = -1$
Both

(A1)
(3)

Alternative; cover up rule

$x = \frac{1}{2} \qquad A = \frac{2\times\frac{1}{2}+3}{2\times\frac{1}{2}+1} \qquad \left(=\frac{4}{2}\right)$

$x = \frac{1}{2}$ and $x = -\frac{1}{2}$ used to find A and B

(M1)

$x = -\frac{1}{2} \qquad B = \frac{2\times(-\frac{1}{2})+3}{2\times(-\frac{1}{2})-1} \qquad \left(=\frac{2}{-2}\right)$
SC NMS

$A = 2 \quad B = -1$
A and B both correct 3 / 3
One of A or B correct 1 / 3

(A1A1)

(3)

Condone poor algebra for M1 if continues correctly.

(b)
$$\begin{array}{r} 3x \\ 4x^2-1\overline{)12x^3-7x-6} \end{array}$$

$12x^3 - \underline{3x}$
$-4x - 6$
Complete division leading to values for C and D

M1

$C = 3$

A1

$D = -2$

A1

C = 3, D = −2 stated or written in expression.
SC B1
C = 3 , D not found or wrong;
D = −2, C not found or wrong.

3

Alternative

$$\frac{12x^3-7x-6}{4x^2-1}=\frac{12x^3-3x-4x-6}{4x^2-1}=3x-\frac{2(2x+3)}{4x^2-1}$$

(M1)

$C = 3$

(A1)

$D = -2$

(A1)

C = 3, D = −2 stated or written in expression.
SC B1
C = 3 , D not found or wrong;
D = −2 , C not found or wrong.

(3)

Alternative

$12x^3 - 7x - 6 = 4Cx^3 - Cx + 2Dx + 3D$

Complete method for C and D

(M1)

$C = 3$

(A1)

$D = -2$

(A1)

C = 3, D = −2 stated or written in expression.
SC B1
C = 3 , D not found or wrong;
D = −2 , C not found or wrong.

(3)

Alternative

$x = 0$ $\quad$ $x = 1$

$6 = -3D \qquad -\frac{1}{3} = C + \frac{5}{3}D$

Use two values of x to set up simultaneous equations

(M1)

$C = 3$

(A1)

$D = -2$

(A1)

C = 3, D = −2 stated or written in expression.
SC B1
C = 3 , D not found or wrong;
D = −2 , C not found or wrong.

(3)

Complete division for M1; obtain a value for $C(Cx)$ and a remainder $ax + b$

(c) $\int 3x - 2\left(\frac{2}{2x-1} - \frac{1}{2x+1}\right) dx$

Use parts (a) and (b) to obtain integrable form

M1

$3\frac{x^2}{2}$

ft on C

A1ft

$-2\left(\ln(2x-1) - \frac{1}{2}\ln(2x+1)\right)$

Both correct; ft on A, B and D
Condone missing brackets

A1ft

$\frac{3}{2}(4-1) - 2\left(\left(\ln 3 - \frac{1}{2}\ln 5\right) - \left(\ln 1 - \frac{1}{2}\ln 3\right)\right)$

Correct substitution of limits

m1

$\frac{9}{2} - 3\ln 3 + \ln 5 = \frac{9}{2} + \ln\left(\frac{5}{27}\right)$

$p = \frac{9}{2} \qquad q = \frac{5}{27}$

A1

Form $\int Cx + \left(\frac{P}{2x-1} + \frac{Q}{2x+1}\right)$ using candidate's P, Q, C for M1.

Condone missing dx.

$\int Cx \,\mathrm{d}x = C\frac{x^2}{2}$ for A1ft

ISW extra terms eg $\frac{12}{4x^2-1}$ for first three terms only; max 3/5.

Candidate's C; must have a value.

$\int \frac{4x+6}{4x^2-1}\,\mathrm{d}x = \int \frac{4x}{4x^2-1} + \frac{6}{4x^2-1}$ dx is an integrable form,

as $\int \frac{1}{x^2-a^2}\,\mathrm{d}x = \frac{1}{2a}\ln\left(\frac{x-a}{x+a}\right)$ is in the formula book, but they **must** try to integrate to show they know this, **or** use partial fractions again with $\frac{6}{4x^2-1} = \frac{3}{2x-1} - \frac{3}{2x+1}$ for M1.

Substitute limits into $C\frac{x^2}{2}$ + $m$ln(2x − 1) + $n$ln(2x + 1), or equivalent, for m1; substitution must be completely correct.

Condone $\frac{9}{2} - \ln\left(\frac{27}{5}\right)$ for A1.

5

[11]

Q20.

$\int \frac{\mathrm{d}y}{y^2} = \int x \sin 3x \,\mathrm{d}x$

Correct separation and notation; condone missing integral signs

B1

$\int \frac{\mathrm{d}y}{y^2} = -\frac{1}{y}$

B1

$\int x \sin 3x \, dx = x\left(-\frac{1}{3}\cos 3x\right)$

$u = x \quad \frac{dv}{dx} = \sin 3x$

$\frac{du}{dx} = 1 \quad v = k\cos 3x$

Use parts with correct substitution into formula

$-\int -\frac{1}{3}\cos 3$

A1

$= -\frac{1}{3}x\cos 3x \, dx + \frac{1}{9}\sin 3x$

CAO

A1

$-\frac{1}{y} = -\frac{1}{3}x\cos 3x + \frac{1}{9}\sin 3x + C$

$-1 = -\frac{1}{3} \times \frac{\pi}{6}\cos\left(\frac{\pi}{2}\right) + \frac{1}{9}\sin\left(\frac{\pi}{2}\right) + C$

Use $x = \frac{\pi}{6}$ $y = 1$ to find C

M1

$C = -\frac{10}{9}$

CAO

A1

$-\frac{1}{y} = -\frac{1}{9}(3x\cos 3x \, dx - \sin 3x + 10)$

And invert to $-y = -\frac{9}{(.....)}$

m1

$y = \frac{9}{3x\cos 3x - \sin 3x + 10}$

CSO, condone first B1 not given

A1

Second M1 finding C; substitute $x = \frac{\pi}{6}$ $y = 1$ into $f(y) = px\cos 3x + q\sin 3x + C$ and evaluate using radians.

Must calculate a value of C.

m1 for reaching form $\pm\frac{k}{y} = \frac{1}{9}(Px\cos 3x\ Q\sin 3x + R)$ where P and Q are ± 3 or $\pm\frac{1}{3}$ or ± 1 and inverting to $\pm\frac{y}{k} = \frac{9}{(Px\cos 3x + Q\sin 3x + R)}$

[9]

Q21.

(a) (i) $\left(x - \frac{3}{2}\right)^2$

or $p = 1.5$ stated

M1

$\left(x - \frac{3}{2}\right)^2 + \frac{11}{4}$

$(x - 1.5)^2 + 2.75$

A1

Mark their final line as their answer

2

(ii) $x = \frac{3}{2}$

correct or FT their "$x = p$"

B1 ✓

1

(b) (i) $x^2 - 3x + 5 = x + 5 \Rightarrow x^2 = 4x$

*eliminating x or y and collecting like terms (condone **one** slip)*
***or** $(y - 5)^2 - 3(y - 5) + 5 = y \Rightarrow y^2 - 14y + 45 = 0$*

M1

$(x \neq 0)$ $\Rightarrow x = 4$

A1

$y = 9$

A1

3

(ii) $\frac{x^3}{3} - \frac{3x^2}{2} + 5x\ (+\ c)$

one of these terms correct

M1

another term correct

A1

all correct (need not have + c)

A1

3

(iii) $[\ ]_0^4 = \frac{4^3}{3} - 3\times\frac{4^2}{2} + 5\times 4$

must have earned M1 in part(b)(ii)

F(their x_B){−F(0)) "correctly sub'd"

M1

$= 17\frac{1}{3}$

$\left(\frac{64}{3} - 24 + 20 =\right)\ \frac{52}{3}$ or $\frac{104}{6}$ *etc*

condone 17.3 but not $16\frac{4}{3}$ *etc*

A1

Area trapezium $\frac{1}{2}(x_B)(5 + y_B)$

FT their numerical values of x_B, y_B

Area = $\frac{1}{2}$ × 4 × 14 (= 28)

B1√

Area of shaded region 28 − $17\frac{1}{3}$

$= 10\frac{2}{3}$

CSO; $\frac{32}{3}$ *, accept 10.7 or better*

A1

4

[13]

Q22.

(a) …… = $\left(x^{\frac{3}{2}}\right)^2 - 2x^{\frac{3}{2}} + 1 = x^3 - 2x^{\frac{3}{2}} + 1$

B2 for $x^3 - 2x^{\frac{3}{2}} + 1$ or $x^3 - 2x\sqrt{x} + 1$
(B1 fully correct unsimplified expression.

seen eg $\left(x^{\frac{3}{2}}\right)^2 - x^{\frac{3}{2}} - x^{\frac{3}{2}} + 1$

or B1 for either $x^3 + 2x^{\frac{3}{2}}$ … OE seen

or $x^3 + 2x^{\frac{3}{2}} - 1$ OE seen)

or B1 for $-x^3 + 2x^{\frac{3}{2}} - 1$ OE seen)

B2,1,0

2

(b) $\int\left(x^{\frac{3}{2}} - 1\right)^2 \mathrm{d}x = \frac{x^4}{4} - \frac{2x^{\frac{5}{2}}}{2.5} + x \ (+c)$

Ft on correct integration of all non $x^{\frac{3}{2}}$ terms
(at least two) in c's expression. in (a)

B1F

Integration of a $k x^{\frac{3}{2}}$ as $\lambda x^{\frac{5}{2}}$ (ie power correct)

M1

{= $0.25x^4 - 0.8x^{2.5} + x \ (+c)$}

Correct integration of c's $x^{\frac{3}{2}}$ term(s)
ACF

A1F

3

(c) $\int_1^4\left(x^{\frac{3}{2}} - 1\right)^2 \mathrm{d}x = \left(\frac{4^4}{4} - \frac{2(4^{\frac{5}{2}})}{2.5} + 4\right) - \left(\frac{1}{4} - \frac{2}{2.5} + 1\right)$

F(4) − F(1) attempted following integration. If F(x)
incorrect, ft c's answer to (b) provided integration attempted

M1

$\{= \frac{212}{5} - \frac{9}{20} = 42.4 - 0.45\} = 41.95$

41.95 OE eg 839 / 20

Since 'Hence' NMS scores 0 / 2

A1

2

[7]

Q23.

(a) (i) (When $x = 2$) $\frac{dy}{dx} = 12 - 1 - 11 = 0$

AG Must see intermediate evaluations

B1

1

(ii) $\frac{4}{x^2} = 4x^{-2}$ {so $\frac{dy}{dx} = 3x^2 - 4x^{-2} - 11$}

$\frac{4}{x^2} = 4x^{-2}$, seen in (a)(ii) or earlier. PI by $\pm 8x^{-3}$ term in answer

B1

$\frac{d^2y}{dx^2} = 6x + 8x^{-3}$

Correct powers of x correctly obtained from differentiating the first two terms

M1

$6x + 8x^{-3}$ ACF

A1

When $x = 2$, $\frac{d^2y}{dx^2} = 12 + 8/8 = 13$

A1

4

(iii) Since $\frac{d^2y}{dx^2} > 0$, P is a minimum point.

Ft on c's value of $y''(2)$ in (a)(ii) but must see reference to sign of $y''(2)$ either explicitly or as inequality, as well as the correct ft conclusion

E1F

1

(b) $\int\left(3x^2-\frac{4}{x^2}-11\right) dx = x^3 + 4x^{-1} - 11x\ (+c)$

Attempt to integrate $\frac{dy}{dx}$ with at least two of the three terms integrated correctly

M1

(y =) $x^3 + 4x^{-1} - 11x(+c)$

For $x^3 + 4x^{-1} - 11x$ OE even unsimplified.

1

When $x = 2$, $y = 1 \Rightarrow 1 = 8 + 2 - 22 + c$

Substituting
$x = 2$, $y = 1$ into $y = F(x) + $ 'c' in attempt to find constant of integration, where $F(x)$ follows attempted integration of expression for $\frac{dy}{dx}$

$y = x^3 + 4x^{-1} - 11x + 13$

ACF

A1

4

[10]

Q24.

(a) (i) $\frac{dy}{dx} = -1 - 4x^3$

one of these terms correct

M1

all correct (no + c)

A1

(When $x = 1$, grad =) –5

(Check that $\frac{dy}{dx}$ is actually correct!)

A1cso

3

(ii) $y - 12$ = 'their grad' $(x - 1)$

any form of equation through (1, 12) and attempt at c if using $y = mx + c$

M1

$y = -5x + 17$ (or $y = 17 - 5x$)

FT their gradient

Condone $y = -5x + c$, $c = 17$ etc

A1√

2

(b) (i) $14x - \frac{x^2}{2} - \frac{x^5}{5}$

one of these terms correct

M1

another term correct

A1

all correct (may have + c)

A1

$[\ ]_{-2}^{1} =$

$\left(14 - \frac{1}{2} - \frac{1}{5}\right) - \left(-28 - 2 + \frac{32}{5}\right)$

F(1) and F(–2) attempted

m1

= 36.9 OE

Condone recovery to this value

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 3 \times 12$

Correct area of triangle unsimplified

M1

= 18

$\Rightarrow$ shaded area = 18.9

A1cso

2

[12]

Q25.

(a) $(1 - x)^3 = 1 - 3x + 3x^2 - x^3$

3 terms correct or 1 (±)3 (±)3 (±)1 seen

M1

All correct

A1

2

(b) $(1 + y)^4 = 1 + 4y + 6y^2 + 4y^3 + y^4$

4 terms correct, accept unsimplified

M1

All 5 terms correct and simplified at some stage

A1

$(1 + y)^4 - (1 - y)^3 =$

$(4y + 3y) + (6y^2 - 3y^2) + (4y^3 + y^3) + y^4$

$=7y + 3y^2 + 5y^3 + y^4$

(as required with $p = 3$ and $q = 5$)

A2 Be convinced as part answer is given

(A1 for three terms found correctly or if found correct values for p and q but did not show $7y + y^4$.)

A2,1

4

(c) $$\int\left[\left(1+\sqrt{x}\right)^4 - \left(1-\sqrt{x}\right)^3\right]\mathrm{d}x =$$

$$\int\left(7\sqrt{x} + 3x + 5x\sqrt{x} + x^2\right)\mathrm{d}x$$

*Use of part (**b**)….$y \to \sqrt{x}$ OE before any integration*

M1

$$\int\left(7x^{0.5} + 3x + 5x^{1.5} + x^2\right)\mathrm{d}x$$

$$= \frac{7x^{1.5}}{1.5} + \frac{3x^2}{2} + \frac{5x^{2.5}}{2.5} + \frac{x^3}{3}(+c)$$

Correct integration of an x^k term where k is non–integer

m1

$$= \frac{14}{3}x^{1.5} + \frac{3}{2}x^2 + 2x^{2.5} + \frac{1}{3}x^3(+c)$$

Coeffs simplified; condone absent (+c)

Ft on c's p and q ie 2nd term $+\frac{p}{2}x^2$ *and 3rd term is* $+\frac{2q}{5}x^{2.5}$
(A1F for three of these four ft term's or for four correct ft terms unsimplified)

A2,1F

4

[10]

Q26.

(a) $y = x + 3 + \frac{8}{x^4} = x + 3 + 8x^{-4}$

For $\frac{8}{x^4} = 8x^{-4}$ *PI by correct differentiation of 3rd term*

B1

$\frac{\mathrm{d}y}{\mathrm{d}x} = 1 - 32x^{-5}$ or $1 - \frac{32}{x^5}$

kx^{-5} *OE*

M1

For either

A1

3

(b) When $x = 1$, $y = 12$

B1

When $x = 1$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 1 - 32 = -31$

Attempt to find value of $\frac{dy}{dx}$ *when* $x = 1$

M1

Tangent: $y - 12 = -31(x - 1)$
*Only ft on c's answer to (**a**). Any correct (ft on c's (a)) form.*

A1F

3

(c) $1 - 32x^{-5} = 0$

$1 - 32x^{-5} = 0$ or c's $\frac{dy}{dx}$ *= 0*

M1

$\Rightarrow x^5 = 32$

Attempt to form x^n = const (≠0). PI by next line

m1

$\Rightarrow x = 2$

CSO

A1

(Coordinates of *M*) (2, 5.5)

CSO

A1

4

(d) (i) $\int\left(x + 3 + \frac{8}{x^4}\right)dx$

$= \frac{x^2}{2} + 3x - \frac{8}{3}x^{-3} + c$

Power −3 correctly obtained

M1

$-\frac{8}{3}x^{-3}$

A1

$\frac{x^2}{2} + 3x + c$

B1

3

(ii)

$= \left(2 + 6 - \frac{1}{3}\right) - \left(\frac{1}{2} + 3 - \frac{8}{3}\right)$

Attempting to calculate F(2) − F(1) where F(x) is c's answer to part (d)(i) provided F is not just the c's integrand (x + 3 + 8/x^4)

M1

$$\frac{9}{2}+\frac{7}{3}=\frac{41}{6}$$

OE Accept 6.83 or better provided d(i) used

A1

2

(e) $k = -5.5$

Ft on $-y_M$ from part (c).

B1F

1

[16]

Q27.

(a) (i) $\int \frac{1}{3+2x} \mathrm{d}x$

$= k \ln (3 + 2x)$

Where k is a rational number

M1

$= \frac{1}{2} \ln(3+2x)+c$

Or

if substitution $u = 3 + 2x$, $du = 2dx$

$\int = \int \frac{1}{u}\frac{du}{2} = k\ln u$ *M1*

$= \frac{1}{2} \ln(3+2x)+c$ *A1*

A1

2

(b) $u = x \quad \mathrm{d}v = \sin\frac{x}{2}$

$\int \sin\frac{x}{2}(dx) = k\cos\frac{x}{2}, \ \frac{d}{dx}(x) = 1$

where k is a constant

M1

$\mathrm{d}u = 1 \quad v = -2\cos\frac{x}{2}$

All correct

A1

$$\int = -2x\cos\frac{x}{2} - \int -2\cos\frac{x}{2}(\mathrm{d}x)$$

***Correct** substitution of their terms into parts formula (watch signs carefully)*

m1

$$= -2x\cos\frac{x}{2} + 4\sin\frac{x}{2} + c$$

CAO

A1

4

[6]

Q28.

(a) (i) $\int \frac{\mathrm{d}x}{\sqrt{x}} = \int \sin\left(\frac{t}{2}\right)\mathrm{d}t$

Correct separation; condone missing integral signs.

B1

$$2\sqrt{x} = -2\cos\left(\frac{t}{2}\right)(+k)$$

$$p\sqrt{x} = q\cos\left(\frac{t}{2}\right)$$

Condone missing + k

M1

$$x = \left(-\cos\left(\frac{t}{2}\right) + C\right)^2$$

Must have previous line correct

A1

3

(ii) (1, 0) $2 = -2 + k$ or $1 = (-1 + C)^2$

Use (1, 0) to find a constant

M1

$k = 4$ or $C = 2$

ft on $C = p - q$ from (a)(i)

A1ft

$$x = \left(2 - \cos\left(\frac{t}{2}\right)\right)^2$$

cso applies to (a)(ii)

A1

3

(b) (i) Greatest height when $\cos(bt) = -1$

M1

Greatest height = 9 (m)

ft is (their a + 1)²

A1ft

2

(ii) $\cos\left(\frac{t}{2}\right) = 2 - \sqrt{5}$

$\cos bt = a - \sqrt{5}$

M1

$t = 2\cos^{-1}\left(2 - \sqrt{5}\right) = 3.6$ (seconds 1dp)

condone 3.6 or better (3.618…..)

A1

2

[10]

Q29.

(a) $\int_{-1}^{1} \left(x^3 - 2x^2 + 3\right) dx$

$$= \left[\frac{x^4}{4} - \frac{2x^3}{3} + 3x\right]_{-1}^{1}$$

one term correct

M1

another term correct

A1

all correct (condone + c)

A1

$= \left(\frac{1}{4} - \frac{2}{3} + 3\right) - \left(\frac{1}{4} + \frac{2}{3} - 3\right)$

'their' F(1) – F(–1)
with (–1)³ etc evaluated correctly
but must have earned M1

B1 ✓

$= 4\frac{2}{3}$

$\frac{14}{3}, \frac{56}{12}$ *etc.*

but combined as single fraction

A1cso 5

(b) Area of $\Delta \left(= \frac{1}{2} \times 2 \times 2\right)$

= 2
PI

B1

Shaded region has area $4\frac{2}{3} - 2$
±their (a) ±their ∆ area

M1

$= 2\frac{2}{3}$

$\frac{8}{3}, \frac{32}{12}$ *etc.*

but combined as single fraction

A1cso 3

[8]

Q30.

(a) $(2 + x^2)^3$

$= [(2)^3] + 3(2)^2(x^2) + 3(2)(x^2)^2 \ [+ (x^2)^3]$

For either (1),3,3,(1) OE unsimplified

or $\begin{pmatrix}3\\1\end{pmatrix} 2^2x^2 + \begin{pmatrix}3\\2\end{pmatrix} 2(x^2)^2$ *OE. PI*

M1

$p = 3(2)^2 = 12$

AG Be convinced. Condone left as $12x^2$

A1

$q = 6$

Accept left as $6x^4$

B1

3

(b) (i) $\int \frac{(2+x^2)^3}{x^4}\,dx = \int x^{-4}(8 + 12x^2 + qx^4 x^6)\,dx$

or $\int\left(\frac{8}{x^4} + \frac{12}{x^2} + q + x^2\right)dx$

Uses (a) and either an indication that $\frac{1}{x^n} = x^{-n}$ *in a product PI or cancelling to get at least 3 correct ft terms*

M1

$\int(8x^{-4} + 12x^{-2} + q + x^2)\,dx$

Ft on c's non–zero q. PI by next line in solution

A1F

$= \frac{8x^{-3}}{-3} + \frac{12x^{-1}}{-1} + qx + \frac{x^3}{3}\{+c\}$

Correct integration of either $8x^{-4}$ or $12x^{-2}$; accept unsimplified

M1

Correct integration of both $8x^{-4}$ and $12x^{-2}$; accept unsimplified coefficients

A1

= .... + $6x$ + $\frac{x^3}{3}$ + c

$$\left(= -\frac{8}{3}x^{-3} - 12x^{-1} + 6x + \frac{x^3}{3} + c\right)$$

For $\text{"6"}\,x + \frac{x^3}{3} + c$ *simplified. The only ft is "6" replaced by c's value for q where q is a non-zero integer.*

B1F

5

(ii) $$\int_1^2 \frac{(2+x^2)^3}{x^4}\,dx = \left\{-\frac{8}{3}(2)^{-3} - 12(2^{-1}) + 6(2) + \frac{2^3}{3}\right\} -$$

$$\left\{-\frac{8}{3}(1)^{-3} - 12(1)^{-1} + 6(1) + \frac{1^3}{3}\right\}$$

Dealing correctly with limits; F(2)−F(1) (must have attempted integration to get F ie c's F is not just the integrand)

M1

$$= \left(-\frac{1}{3} - 6 + 12 + \frac{8}{3}\right) - \left(-\frac{8}{3} - 12 + 6 + \frac{1}{3}\right)$$

$$= 16\frac{2}{3}$$

*OE **exact** answer eg 50/3.*
NMS scores 0

A1

2

[10]

Q31.

(a) (i) $$\left(\frac{dx}{d\theta} =\right) - 6\sin 2\theta, \quad \left(\frac{dy}{d\theta} =\right) - 2\sin\theta$$

$$\left(\frac{dx}{d\theta} =\right) p\sin\theta \ \text{ or } \ r\sin\theta\cos\theta$$

$$\left(\frac{dy}{d\theta} =\right) q\sin\theta$$

M1

Both correct.

A1

$$\frac{dy}{dx} = \frac{-2\sin\theta}{-6\sin 2\theta}$$

Use chain rule $\dfrac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$ *; condone one slip*

M1

$$= \frac{2\sin\theta}{6\times 2\sin\theta\cos\theta} = \frac{1}{6\cos\theta}$$

k = 6 must come from correct working seen ***AG***

A1

4

(ii) $\theta = \frac{\pi}{3}$ $\quad m_T = \frac{1}{3}$

ft on k $\left(\dfrac{1}{k \times \frac{1}{2}}\right)$

k need not be numerical

B1ft

$m_N = -3$

ft on m_T

B1ft

$$(x, y) = \left(-\frac{3}{2}, 1\right)$$

B1

Normal $y - 1 = -3\left(x + \frac{3}{2}\right)$

CAO; any correct form, ISW.

$2y + 6x + 7 = 0$

B1

4

(b) $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

p + qcos2x ; Allow different letters for x or mixture eg θ even for A1and the following A1ft

M1

A1

$\int p \, dx = px \quad \int q \cos 2x = \frac{1}{2} q \sin 2x$

Both integrals correct;

ft on p and q

A1ft

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \left[\frac{x}{2} - \frac{1}{4}\sin 2x\right]$$

$$= \left(\frac{\pi}{8} - \frac{1}{4}\right) - \left(-\frac{\pi}{8} - \left(-\frac{1}{4}\right)\right)$$

Correct use of limits;

$F\left(\frac{\pi}{4}\right) - F\left(-\frac{\pi}{4}\right)$ *or* $2F\left(\frac{\pi}{4}\right)$

$F(x) = px + r\sin 2x$ *and* $\sin\frac{\pi}{2}, \sin\left(-\frac{\pi}{2}\right)$

must be evaluated correctly for m1

m1

$= \frac{\pi}{4} - \frac{1}{2}$

CSO OE ISW

A1

5

Alternative

$\int \sin^2 x dx = -\sin x \cos x - \int -\cos x \cos x dx$

Use parts; condone sign slips

(M1)

$$= -\sin x \cos x + \int 1 - \sin^2 x \, dx$$

Use $\cos^2 x = 1 - \sin^2 x$

(m1)

$$2\int \sin^2 x \, dx = -\sin x \cos x + x$$

(A1)

$$2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x \, dx = G\left(\frac{\pi}{4}\right) - G\left(-\frac{\pi}{4}\right)$$

Correct use of limits

(m1)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x \, dx = \frac{\pi}{4} - \frac{1}{2}$$

(A1) **(5)**

[13]

Q32.

(a) $\sqrt{x^5} = x^{\frac{5}{2}}$

Accept k = 2.5

B1 **1**

(b) $\int\left(7\sqrt{x^3} - 4\right)dx = \frac{7}{3.5}x^{3.5} - 4x(+c)$

Index 'k' raised by 1 in integrating x^k

M1

1st term correct follow through on non- integer k

A1F

For $-4x$ as integral of -4

B1 **3**

(c) $y = 2x^{3.5} - 4x + c$ (*)

y = c's answer to (b) with '+ c'
('y =' PI by next line)

B1F

When $x = 1$, $y = 3 \Rightarrow 3 = 2 - 4 + c$

Subst. (1, 3) in attempt to find constant of integration

M1

$y = 2x^{3.5} - 4x + 5$

*Accept c = 5 after correct eqn * which must include 'y ='*
Coefficients must be tidied

A1

3

[7]

Q33.

(a) $10x^2 + 8 = 2(x + 1)(5x - 1) +$

A and B terms correct

M1

$A(5x - 1) + B(x + 1)$

A1

$x = -1 \quad x = \frac{1}{5}$

$A = -3 \quad B = 7$

Use two values of x to find A and B, or set up and solve
$8 + 5A + B = 0$
$-2 - A + B = 8$

SC1 NWS A & B correct $^4/_4$

SC2 NWS A or B correct $^1/_4$

m1A1

4

(b) $$\int \frac{10x^2+8}{(x+1)(5x-1)}\,dx = \int 2 - \frac{3}{x+1} + \frac{7}{5x-1}\,dx$$

Use the partial fractions

M1

$= 2x + C$

B1

a ln (x + 1) + b ln (5x – 1)
condone missing brackets

M1

$$-3\ln(x+1) + \frac{7}{5}\ln(5x-1)$$

F on A and B

A1F

4

[8]

Q34.

$$\int y\,dy = \int \cos\left(\frac{x}{3}\right)dx$$

Separate; condone missing integral signs.

B1

$$\frac{1}{2}y^2 = 3\sin\left(\frac{x}{3}\right) + (C)$$

Accept $\dfrac{\sin\left(\frac{x}{3}\right)}{\frac{1}{3}}$

B1B1

$$\left(\frac{\pi}{2}, 1\right) \quad \frac{1}{2} = 3\sin\frac{\pi}{6} + C$$

Use $\left(\frac{\pi}{2}, 1\right)$ to find C

must be in form $py^2 = q \sin\left(\frac{x}{3}\right) + C$

M1

$C = -1$

A1F

$y^2 = 6\sin\left(\frac{x}{3}\right) - 2$

CSO

A1

[6]

Q35.

(a) $p = -3$; $q = 3$

Accept even if just embedded in the expansion

B1;B1 2

(b) (i) $\int\left(1-\frac{1}{x^2}\right)^3 \, dx = \int\left(1 - 3x^{-2} + 3x^{-4} - x^{-6}\right)dx$

Uses (a) with indication of integration and

indication of $\frac{1}{x^n} = x^{-n}$ *PI*

M1

$= x + 3x^{-1} - x^{-3} + \frac{1}{5}x^{-5} \; \{+c\}$

At least three powers of x *correctly obtained*

m1

Ft on c's non-zero integers p *and* q*.*
A1F if 3 of the 4 terms are correct (ft) or
if all correct (ft) but left unsimplified
Condone missing +c*.*

A2F,1F 4

(ii) $\int_{\frac{1}{2}}^{1}\left(1-\frac{1}{x^2}\right)^3 \, dx = \left(1 + 3 - 1 + \frac{1}{5}\right) - \left(\frac{1}{2} + 6 - 8 + \frac{32}{5}\right)$

Attempting to calculate F(1) – F(1/2)
where F is c's answer to part (b)(i)
provided F is not the integrand or the c's

equivalent of the integrand $(1-\frac{1}{x^2})^3$.

M1

$= -\frac{17}{10}$

OE exact answer eg –1.7

A1

2

[8]